Curriculum Vitae of Ilyas Fatkhullin

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Positions

Stanford University, USA 10/2026 – Present
Postdoctoral Scholar, Department of Management Science and Engineering
Mentors: Prof. Renyuan Xu and Prof. Madeleine Udell

Education

ETH Zurich, Switzerland 12/2021 – 09/2026
Ph.D. in Computer Science, ETH AI Center Fellow
Advisor: Prof. Niao He
Research visit: ***Georgia Institute of Technology***, USA (with Prof. Guanghui Lan), 03–09/2025
Dissertation Title: Structure, Geometry, and Noise in Stochastic First-Order Optimization

Technical University of Munich, Germany 09/2020 – 11/2021
M.Sc. in Mathematics
External Advisor: Prof. Peter Richtárik Local Advisor: Prof. Michael Ulbrich
Thesis Topic: Error Compensation Method for Communication Efficient Distributed Training

Moscow Institute of Physics and Technology, Russia 09/2016 – 08/2020
Advisor: Prof. Boris Polyak
Thesis Topic: Optimization Landscape of Linear Quadratic Regulator Problem with Output Feedback

Publications (Google Scholar)

Refereed Journal Publications

[1] I. Fatkhullin, F. Hübler, G. Lan. *Can SGD Handle Heavy-Tailed Noise?*
Accepted to SIAM Journal on Optimization, 2026. Preprint: arXiv:2508.04860. 2025.
- Oral Presentation Award at NeurIPS'25 Workshop on "Optimization for Machine Learning".

[2] I. Fatkhullin, N. He, Y. Hu. *Stochastic Optimization under Hidden Convexity.*
SIAM Journal on Optimization, 2025.

[3] J. Wu, A. Barakat, I. Fatkhullin, N. He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence.*
SIAM Journal on Control and Optimization, 2025 (preliminary version at CDC, 2023).

[4] I. Fatkhullin, I. Sokolov, E. Gorbunov, Z. Li, P. Richtárik. *EF21 with Bells & Whistles: Six Algorithmic Extensions of Modern Error Feedback.*
Journal of Machine Learning Research, 2025.

[5] I. Fatkhullin, B. Polyak. *Optimizing Static Linear Feedback: Gradient Method.*
SIAM Journal on Control and Optimization, 2021.
- Cited over 150 times.

[6] B. Polyak., I. Fatkhullin. *Use of Projective Coordinate Descent in the Fekete Problem.*
Computational Mathematics and Mathematical Physics, 2020.

Under Review in Journals

[1] C. Tsai, I. Fatkhullin, L. Zhang, N. He. *Lower Bounds for Anytime Acceleration of Gradient Descent.*
Preprint: arXiv:2607.02053, 2026.

[2] I. Fatkhullin, N. He, G. Lan, F. Wolf. *Global Solutions to Non-Convex Functional Constrained Problems with Hidden Convexity.* Preprint: arXiv:2511.10626, 2025.
- Oral Presentation Award at NeurIPS'25 Workshop "Constrained Optimization for Machine Learning".

Refereed Machine Learning Conference Publications

[1] F. Wolf, I. Fatkhullin, N. He. *Global Optimality for Constrained Exploration via Penalty Regularization.* NeurIPS, 2026.
- Oral Presentation Award at the European Workshop on Reinforcement Learning (EWRL'26).

[2] A. Sadiev, P. Richtárik, I. Fatkhullin. *Second-order Optimization under Heavy-Tailed Noise: Hessian Clipping and Sample Complexity Limits.* NeurIPS, 2025.

[3] F. Sun, I. Fatkhullin, N. He. *Natural Gradient VI: Guarantees for Non-Conjugate Models.* NeurIPS, 2025.

[4] R. Islamov, Y. As, I. Fatkhullin. *Safe-EF: Error Feedback for Nonsmooth Constrained Optimization.* ICML, 2025.

[5] F. Hübler*, I. Fatkhullin*, N. He. *From Gradient Clipping to Normalization for Heavy-Tailed SGD.* AISTATS, 2025. *Equal contribution.

[6] I. Fatkhullin, N. He. *Taming Nonconvex Stochastic Mirror Descent with General Bregman Divergence.* AISTATS, 2024.

[7] I. Fatkhullin, A. Tyurin, P. Richtárik. *Momentum Provably Improves Error Feedback!* NeurIPS, 2023.

[8] J. Yang, X. Li, I. Fatkhullin, N. He. *Two Sides of One Coin: the Limits of Untuned SGD and the Power of Adaptive Methods.* NeurIPS, 2023.

[9] A. Barakat, I. Fatkhullin, N. He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space.* ICML, 2023.

[10] I. Fatkhullin, A. Barakat, A. Kireeva, N. He. *Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies.* ICML, 2023.

[11] P. Richtárik, I. Sokolov, I. Fatkhullin, E. Gasanov, Z. Li, E. Gorbunov. *3PC: Three Point Compressors for Communication-Efficient Distributed Training and a Better Theory for Lazy Aggregation.* ICML, 2022.
- Spotlight Presentation Award at ICML Main Track (top 5% submissions).

[12] I. Fatkhullin*, J. Etesami*, N. He, N. Kiyavash. *Sharp Analysis of Stochastic Optimization under Global Kurdyka-Lojasiewicz Inequality.* NeurIPS, 2022. *Equal contribution.

[13] P. Richtárik, I. Sokolov, I. Fatkhullin. *EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback.* NeurIPS, 2021.
- Oral Presentation Award at NeurIPS Main Track (top 1% submissions).
- Cited over 300 times.

Fellowships and Awards

Top Reviewer Award by NeurIPS'26 — 10/2026

Swiss National Science Foundation (SNSF) Postdoc.Mobility Fellowship — 10/2026
Project: *Structure Exploiting Optimization for Data-Efficient Reinforcement Learning and Games*
Host: Stanford University

Oral Presentation Award from EWRL'26 Workshop — 10/2026

Gold Reviewer Award by ICML'26 — 05/2026

President's Postdoctoral Fellowship Program, Georgia Institute of Technology — 04/2026
Competitive fellowship offer, declined in favor of SNSF

Selected for Rising Stars Workshop in Industrial Engineering IOE-ISyE-MS&E — 03/2026
Event jointly organized by the UMichigan, Georgia Tech and Stanford

Oral Presentation Award from OPT 2025 Workshop at NeurIPS'25 — 11/2025

Oral Presentation Award from COML 2025 Workshop at NeurIPS'25
for the *Best fundamental (theoretical or algorithmic) contribution* 10/2025

TMLR Expert Reviewer 09/2025

Rising Star in AI Award, KAUST 02/2023

ICCOPT 2022 Travel Grant 05/2022

Spotlight Presentation Award at ICML'22 07/2022

Oral Presentation Award at NeurIPS'21 12/2021

ETH AI Center Doctoral Fellowship 02/2021

German Academic Exchange Service (DAAD) Scholarship for MSc in Germany 04/2020

PreDoc Program Fellowship in Mathematics, Technical University of Munich 03/2020

Teaching Experience

Lecturer, "Data Analysis and Machine Learning", Ashesi University, Ghana 05/2025
Part of the Ashesi–ETH Master in Mechatronic Engineering.

- Independently designed and delivered a 3-week intensive graduate-level course, including syllabus, lectures, coding labs, assignments, a final project, and exam.
- Student evaluations (26/34 response rate): overall instructor rating 4.45/5.00.
- Received particularly strong scores in preparation of lessons (4.54), encouraging participation (4.54), accessibility and approachability (4.58), and use of instructional technology (4.50).

Teaching Assistant, "Optimization for Data Science", ETH Zurich Spring 2023, 2024

- Led weekly tutorials in a core M.Sc.-level optimization course with 100+ students.
- Co-designed homework assignments and contributed exam questions for midterm and final assessments.
- Held office hours and provided individualized support for students.

Instructor, Seminar "Advanced Topics in Machine Learning", ETH Zurich Fall 2023, 2024

- Guided research-paper discussions, coached students on presentations and scientific writing.
- Evaluated student presentations and advised on short research exploration projects.

Supervision and Mentoring

Since 2022, I have mentored ten Master's, semester, and research-project students as a primary supervisor. These collaborations have led to publications in prestigious venues (NeurIPS, ICML, AISTATS, SIAM Journal), and mentees have progressed to Ph.D. programs at ETH Zurich, Caltech, Max Planck Institute.

Ph.D. Project Supervision

- Rustem Islamov (University of Basel) — Ph.D. project.
- Yarden As (ETH Zurich) — Ph.D. project.

Master's Theses

- Chung-En Tsai (ETH Zurich, 2026) — subsequently Ph.D. student, ETH Zurich.
- Fangyuan Sun (ETH Zurich, 2026) — subsequently Quant Researcher, WorldQuant.
- Florian Wolf (TU Darmstadt, 2025) — subsequently Ph.D. student, Caltech.
- Jiduan Wu (ETH Zurich, 2023) — subsequently Ph.D. student, Max Planck ETH CLS.
- Harish Rajagopal (ETH Zurich, 2022) — subsequently Software Developer, PQFORCE (Switzerland).

Research Assistants and Interns

- Florian Hübler (ETH Zurich, 2024) — subsequently Ph.D. student, ETH Zurich.

Semester Projects

- Lucas Whitfield (ETH Zurich, 2026) — continued to Master's Thesis at ETH Zurich.
- Fangyuan Sun (ETH Zurich, 2025) — continued to Master's Thesis under my supervision at ETH Zurich.

Research Internships and Short-term Visits

Amazon, Luxembourg 10/2025-02/2026
Applied Scientist Intern in Forecasting & Optimization Team
Worked on large-scale capacity management problems of Amazon's delivery network.

Georgia Institute of Technology, Atlanta, USA
H. Milton Stewart School of Industrial and Systems Engineering 03–09/2025
with Prof. Guanghui (George) Lan
Studied properties of stochastic gradient descent under infinite variance.

King Abdullah University of Science and Technology, Saudi Arabia (remote) 03–09/2021
with Prof. Peter Richtárik
Worked on federated learning algorithms with momentum and quantization.

École Polytechnique Fédérale de Lausanne, Switzerland (remote) 06–10/2020
with Prof. Sebastian Stich and Prof. Martin Jaggi
Studied accelerated methods for convex optimization.

German Electron Synchrotron (DESY), Hamburg; with Prof. Judith Katzy 07–09/2019
Applied adversarial learning methods to the detection of events in high-energy physics.

Helmholtz-Zentrum Berlin, Germany; with Prof. Ji Li 07–08/2018
Developed numerical solvers for physical simulations in materials science.

Professional Activities

Journal Reviewer: Mathematical Programming; Operations Research; SIAM Journal on Optimization; SIAM Journal on Control and Optimization; Journal of Machine Learning Research; IEEE Transactions on Automatic Control; Transactions on Machine Learning Research.

Conference Reviewer: ICML (2022-2025); AISTATS 2023; NeurIPS (2023-2026); ICLR 2024.

Workshop Organizer: Swiss Optimization Symposium, Monte Verità Ascona, Switzerland 08/2026

Session Organizer at SIAM Optimization Conference 06/2026
Location: Edinburgh, United Kingdom
Session Title: "Optimization under Data-driven Constraints"

Session Organizer at INFORMS Optimization Society Conference 03/2026
Location: Atlanta, USA
Session Title: "Methods for Multi-stage Decision Making"

Session Organizer at International Conference on Continuous Optimization 07/2025
Location: University of Southern California, Los Angeles, USA
Cluster and Session Title: "Optimization for Data Science"; "Adaptive Methods in Optimization"

Session Organizer at INFORMS Optimization Society Conference 07/2024
Location: Rice University, Houston, USA
Cluster and Session Title: "Optimization in Data Science"; "Recent Advances in Min-Max Optimization"

Selected Talks

I. Fatkhullin. *Global Optimization under Hidden Convexity: Non-convex Functional Constraints* 06/2026
SIAM Optimization Conference, University of Edinburgh, United Kingdom

I. Fatkhullin. *Can SGD Handle Heavy-Tailed Noise?* 05/2026
Lund University, Lund, Sweden

I. Fatkhullin. *Toward Dependable Reinforcement Learning: Hidden Structures and Safe Optimization* 02/2026
Yale FDS Seminar, USA

I. Fatkhullin. *Natural Gradient Variational Inference: Guarantees for Non-conjugate Models.* 11/2025
SlidesLive NeurIPS 2025, San Diego, USA

I. Fatkhullin. *Can SGD Handle Heavy-Tailed Noise?* 08/2025
H. Milton Stewart School of ISyE, Georgia Institute of Technology, Atlanta, USA

I. Fatkhullin. *Safe Error Feedback with Applications in Humanoid Robot Fleet Training.* 08/2025
International Conference on Stochastic Programming (ICSP 2025), Paris, France

I. Fatkhullin. *Nonconvex Stochastic Mirror Descent with Implications in Machine Learning.* 07/2025
International Conference on Continuous Optimization (ICCOPT 2025), Los Angeles, USA

I. Fatkhullin. *Heavy-Tailed Gradients in AI Models.* 12/2024
ETH AI Center Seminar, Zurich, Switzerland

I. Fatkhullin. *Stochastic Optimization under Hidden Convexity.* 07/2024
European Conference on Advances in Continuous Optimization (EUROPT 2024) , Lund, Sweden

I. Fatkhullin. *Momentum Provably Improves Error Feedback!* 12/2023
SlidesLive NeurIPS 2023, New Orleans, USA

I. Fatkhullin. *Stochastic Policy Gradient Methods for Fisher-non-degenerate Policies.* 07/2023
SlidesLive ICML 2023, Honolulu, USA

I. Fatkhullin. *Policy Gradient Methods in Reinforcement Learning.* 02/2023
Rising Star in AI Symposium, KAUST, Saudi Arabia

I. Fatkhullin. *Stochastic Optimization under Kurdyka-Lojasiewicz Condition.* 06/2022
ETH AI Center Doctoral Seminar, Zurich, Switzerland

I. Fatkhullin. *Practical Algorithmic Extensions of Modern Error Feedback.* 10/2021
Federated Learning One World (FLOW) Seminar

References

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